

Quasi-Newton Yöntemi

$$X_{k+1} = X_k - G_k H_k^{-1}$$

X_0 başlangıç noktası ve H_0 verilsin.

Repeat for $k = 0, 1, 2, \dots$

$$G_k = \nabla F(X_k)$$

$$P_k = -H_k * G_k$$

Set

$$X_{k+1} = X_k + s P_k$$

$$Y_k = g_{k+1} - g_k$$

$$\delta_k = x_{k+1} - x_k$$

$$Y_k * H_{k+1} = \delta_k \text{ (Quasi-Newton Koşulu)}$$

Till

$\|G(X_{k+1})\|$ yeterince küçük oluncaya kadar.

Örnek:

$$f(x) = x_1^2 + \frac{1}{2}x_2^2 + 3 \quad X_0 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad H_0 = I_2$$

Quasi-Newton ile çöz $\delta = ?$ $\gamma = ?$

$$Y_k = g_{k+1} - g_k \quad g = \begin{bmatrix} 2x_1 \\ x_2 \end{bmatrix}$$

$$\delta_k = X_{k+1} - X_k$$

$$X_{k+1} = X_k + s P_k$$

$$P_k = -H_k \cdot g_k$$

$$P_0 = -H_0 \cdot g_0 = - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2x_1 \\ x_2 \end{bmatrix}$$

$$P_0 = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$$

$$X_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + s \cdot \begin{bmatrix} -2 \\ -2 \end{bmatrix} \quad X_1 = \begin{bmatrix} 1 - 2s \\ 2 - 2s \end{bmatrix}$$

$$P_k^T \cdot g_{k+1} = 0 \quad g_1 = \begin{bmatrix} 2x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 - 4s \\ 2 - 2s \end{bmatrix}$$

$$\begin{bmatrix} -2 \\ -2 \end{bmatrix}^T \cdot \begin{bmatrix} 2 - 4s \\ 2 - 2s \end{bmatrix} = 0$$

$$(-2 \cdot (2 - 4s)) + (-2 \cdot (2 - 2s)) = 0$$

$$-4 + 8s - 4 + 4s = 0$$

$$-8 + 12s = 0$$

$$s = \frac{8}{12} = \frac{2}{3}$$

X_1 Hesabı

$$X_1 = X_0 + sP_0$$

$$X_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \frac{2}{3} \cdot \begin{bmatrix} -2 \\ -2 \end{bmatrix}$$

$$X_1 = \begin{bmatrix} 1 - \frac{4}{3} \\ 2 - \frac{4}{3} \end{bmatrix} \quad X_1 = \begin{bmatrix} -\frac{1}{3} \\ \frac{2}{3} \end{bmatrix}$$

δ_0 Y_0 Hesabı

$$Y_0 = g_1 - g_0$$

$$Y_0 = \begin{bmatrix} -\frac{2}{3} \\ \frac{2}{3} \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -\frac{2}{3} - 2 \\ \frac{2}{3} - 2 \end{bmatrix} = \begin{bmatrix} -\frac{8}{3} \\ -\frac{4}{3} \end{bmatrix}$$

$$\delta_0 = \begin{bmatrix} -\frac{1}{3} \\ \frac{2}{3} \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{3} - 1 \\ \frac{2}{3} - 2 \end{bmatrix} = \begin{bmatrix} -\frac{4}{3} \\ -\frac{4}{3} \end{bmatrix}$$